IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-319

1. Topic (12 words max)

CinnamonSri: A Smart IoT-Integrated System for Enhancing Ceylon Cinnamon Production and Export Efficiency

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-319
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Sri Lanka, renowned globally for its premium Ceylon cinnamon, plays a vital role in the international spice market. Despite this reputation, the cinnamon production process still relies heavily on traditional practices. Farmers face challenges such as inconsistent drying due to manual methods, lack of proper grading systems, price uncertainty, and poor disease management. These issues reduce product quality and yield, ultimately affecting export competitiveness and profitability.

One of the most critical issues is maintaining consistent drying conditions. Over-drying or under-drying leads to degradation in quality. Moreover, cinnamon grading is mostly performed manually by trained experts, which limits access and scalability. Pricing is often determined by middlemen, without giving farmers real-time data or predictive power to understand market trends. Additionally, crop diseases go unnoticed until severe, due to the lack of proper early detection tools.

There is a clear gap in the use of modern technology, such as IoT, artificial intelligence, and mobile accessibility, in cinnamon farming. Existing technological solutions are either generic to agriculture or unaffordable for smallholder farmers.

CinnamonSri aims to address these limitations by creating an integrated system using smart IoT devices, machine learning models, and user-friendly mobile applications. The project empowers cinnamon farmers with tools to monitor drying conditions, predict grade and pricing, identify diseases early, and ultimately make data-driven decisions. By introducing automation, AI, and localized interfaces, CinnamonSri will elevate the efficiency, quality, and economic return of cinnamon production in Sri Lanka.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution, CinnamonSri, consists of four tightly integrated technology components tailored for the Ceylon cinnamon industry:

Smart Drying Automation (IoT):

A DHT22 sensor system is used to measure real-time temperature and humidity inside the drying chamber. An ESP8266 microcontroller activates a fan and heater through a relay based on sensor values. Data is uploaded to Firebase and visualized in a mobile app that offers Sinhala voice alerts, allowing farmers to monitor drying status remotely and maintain export-grade quality.

AI-Powered Quality & Price Prediction:

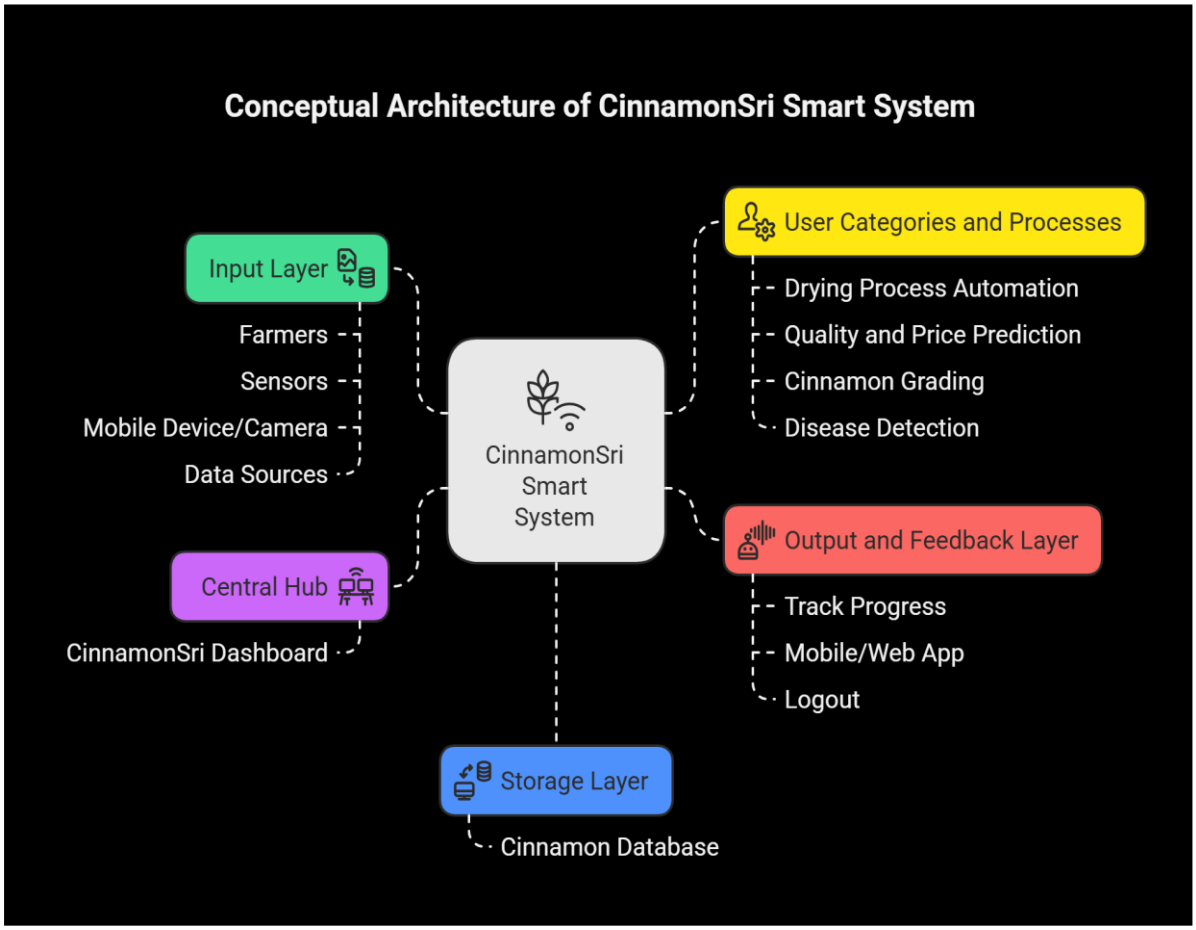
Farmers enter values such as weight before/after drying and other attributes into a mobile/web app. An ML model predicts the cinnamon quality grade (e.g., Premium, Grade A) and estimated price per kilogram, empowering informed selling decisions.

Image-Based Cinnamon Grading:

Farmers or collectors can upload cinnamon strip images. Using ML and computer vision techniques, the system analyzes visual features (e.g., curl, thickness, color) and predicts the grade. This generates a detailed PDF report for export purposes also.

Disease Detection Module:

- Farmers upload or capture images of **cinnamon leaves or bark** via a mobile app.
- The app sends the image to a CNN (Convolutional Neural Network)-based model trained on cinnamon disease datasets.
- The model classifies the image and identifies if **leaf spot** or other diseases are present.
- Results and suggested treatments (based on expert knowledge) are sent back instantly.
- A **Sinhala/Tamil voice alert** can notify farmers with the disease name and action needed.
- Data is also logged into a central dashboard for long-term monitoring and learning.


Conceptual Architecture of CinnamonSri Smart System
Input Layer
Farmers
Sensors
Mobile Device/Camera
Data Sources
Central Hub
CinnamonSri Dashboard
CinnamonSri Smart System
User Categories and Processes
Drying Process Automation
Quality and Price Prediction
Cinnamon Grading
Disease Detection
Output and Feedback Layer
Track Progress
Mobile/Web App
Logout
Storage Layer
Cinnamon Database

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project requires interdisciplinary knowledge in IoT, machine learning, embedded systems, and agricultural processes specific to cinnamon.

- IoT and Embedded Systems: Developing a smart drying system involves understanding temperature/humidity sensors (DHT22), microcontroller programming (ESP8266/Arduino), and relay-controlled hardware. Firebase integration ensures cloud-based real-time updates.
- Machine Learning and Data Science: Creating reliable prediction models for grading and pricing requires supervised learning techniques like Random Forest, SVM, and XGBoost. Feature engineering from real data (e.g., moisture content, curl, weight) is critical.
- Image Processing: Grading and disease detection tools require image capture, preprocessing (e.g., filtering, feature extraction), and training of deep learning models like CNNs. Knowledge of OpenCV and TensorFlow/Keras is essential.
- Agricultural Domain Expertise: Familiarity with cinnamon grading criteria (ISO 6539:2014), drying standards, and common disease symptoms is required. This includes collaborating with the Department of Export Agriculture or local cinnamon cooperatives.
- Data Collection: Historical cinnamon data (e.g., quality vs. price), images of graded strips, and disease-affected samples will be sourced through partnerships with farmers and research institutions. When insufficient, data augmentation techniques or simulated datasets will be employed.
- Mobile & Web Development: User interfaces must be developed using technologies like React Native or Flutter with local language support (Sinhala/Tamil) and voice notifications. App performance optimization is crucial for rural deployment.

This combination of technical and contextual expertise ensures CinnamonSri is both cutting-edge and grounded in local agricultural realities.

8. Objectives and Novelty

Main Objective

To design and develop CinnamonSri – a smart, integrated IoT and AI-based system that enhances the quality, consistency, and economic value of Ceylon cinnamon production by automating drying processes, predicting product grading and pricing, generating professional quality reports, and detecting crop diseases to support smallholder farmers and exporters in Sri Lanka.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22180834 Sandunmeth M.M.S	**Smart IoT-Based Cinnamon Drying System**	1. Integrate DHT22 sensor to collect real-time temperature and humidity data. 2. Connect heater and fan to the system using a relay module for automated control. 3. Program a microcontroller (Arduino/ESP8266) to analyze sensor data and switch heater/fan ON or OFF accordingly. 4. Send data to Firebase Realtime Database for remote access. 5. Develop a React Native mobile app to display drying status	Real-time sensor-based automated drying for cinnamon – rarely seen in local agriculture.

		6. Test the system inside a drying box with real cinnamon samples and evaluate temperature stability.	
IT22230638 Nathasha K.D.S	**AI/ML-Based Cinnamon Quality and Price Prediction Tool**	1. Collect and preprocess cinnamon quality and market price data. 2. Develop AI/ML model for quality classification and price prediction. 3. Create user-friendly mobile/web interface for farmer inputs. 4. Deploy model on cloud or local devices for accessibility.	• Farmer-centric AI tool using simple inputs for quality grading and localized price prediction, reducing dependency on costly IoT hardware.
IT22101242 Tharuka M.S	To develop an AI-based system that captures an image of cinnamon strips, identifies key grading features, predicts the correct grade using machine learning, and generates a detailed PDF report explaining the grade and its industrial use	1. Capture or upload images of cinnamon strips. 2. Preprocess images and extract features (color, length, texture, curl). 3. Train and use a machine learning model to predict the cinnamon grade. 4. Generate a PDF report that includes: • Detected features • Predicted grade • Explanation of why that grade was given	• Combines computer vision and ML to perform automated grading of cinnamon (a task usually done manually by experts). • Generates a smart, auto-filled PDF report explaining the grading decision and usage context. • Enables fast, consistent, and digital report of cinnamon grades using just an image, improving

		Usage of that grade in the industry	efficiency in the spice industry
IT22112200 Iruppuge R.A.F	**Detect leaf spot & other diseases early using image classification**	1. Collect & label cinnamon leaf disease images (focus on rainy season and leaf spot) 2. Preprocess (resize, filter, enhance images) 3. Train CNN model (e.g., using TensorFlow) 4. Integrate prediction with CinnamonSri dashboard 5. Show prevention/treatment via app	• First cinnamon-specific AI model for leaf spot detection • Works in rainy season with real-time image analysis • Helps farmers take quick action before disease spreads • Sinhala voice alerts improve accessibility • Low-cost mobile camera-based tool without expensive hardware

1. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22180834 Sandunmeth M.M.S	This member's component focuses on IoT-based automation, which aligns with the IT specialization under embedded systems, real-time data communication, and smart agriculture solutions. By integrating sensors, microcontrollers, Firebase, and a mobile app, the system applies core IT skills to address real-world agricultural challenges.
IT22230638 Nathasha K.D.S	This member is working on AI/ML-driven analytics, focusing on building predictive models for cinnamon grading and pricing. It aligns with the IT specialization through data analysis, machine learning model development, and cloud-based deployment. It supports decision-making in agriculture using IT techniques like regression/classification and user-friendly app design.
IT22101242 Tharuka M.S	This component applies computer vision and image processing techniques to automate cinnamon grading, which directly complies with the IT specialization in AI, image recognition, and intelligent system development. The creation of dynamic PDF reports also demonstrates knowledge in software engineering and report automation tools.
IT22112200 Iruppuge R.A.F	This component aligns with the IT specialization by applying artificial intelligence through deep learning to classify cinnamon leaf diseases. It uses image processing to detect specific patterns like leaf spots. The system is integrated with a mobile app and farmer-friendly dashboard to make it easy to use in the field. It also acts as a decision support tool by giving instant suggestions for managing the disease.

2. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.